

Module 1 In Class Activity

Module one starts with selecting and configuring a Java programming environment using JetBrains IntelliJ IDEA, Visual Studio, or similar IDEs. With any text editor it is needed to install JDK, this is a .exe file installed in a computer which provides the required data to compile Java code and run programs inside IDEs.

Java Anatomy refers to the structure of a Java program, at the start of each program first line “public class ...” determines the class name which should be the same as the .java file name. Inside this class’s braces will be the main body, which will contain the main code of our program and will normally look like “public class void main(String[] args),” similar to ‘int main()’ used in C++.

Tables and Chairs

```
1 // File Name: TableAndChairs.java
2 // Student Starter Code with TODOs
3
4 public class TablesAndChairs {
5
6     public static void main(String[] args) {
7
8         // TODO 1: Use System.out.println() statements to display the table and chairs pattern.
9         System.out.println("X");
10        System.out.println("X");
11        // TODO 2: Make sure the spacing and alignment match the required pattern.
12        System.out.println("X                XXXXXXXXXX X");
13        System.out.println("XXXXX          X          X          XXXXXX");
14        System.out.println("X  X  X  X  X  X  X  X  X  X");
15        System.out.println("X  X  X  X  X  X  X  X  X  X");
16        // TODO 3: Compile and run the program to confirm the output displays correctly.
17
18        System.out.println("TODO: Replace this line with the table and chairs pattern.");
19    }
20 }
```

A simple program to demonstrate the output structure in console using Java. With the use of *system.out.printl* the program will output characters between double quotes in the screen. Many blank spaces were added to these outputs to demonstrate how even these are printed in some form in the console, changing the structure in which can be perceived. This same technique may be used to align the output, separate lines, or print calculations.

Rectangle

```
1 // File Name: Rectangle.java
2 // Student Starter Code with TODOs
3
4 public class Rectangle {
5
6     // TODO 1: Declare two private double attributes: width and height.
7     private double width; 4 usages
8     private double height; 4 usages
9
10    // TODO 2: Create a constructor that accepts width and height as parameters.
11    public Rectangle(double width, double height) { 1 usage
12        this.width = width;
13        this.height = height;
14    }
15
16    // TODO 3: Create a method named calculateArea() that returns width * height.
17    public double calculateArea() { 1 usage
18        return width * height;
19    }
20
21    // TODO 4: Create a method named calculatePerimeter() that returns 2 * (width + height).
22    public double calculatePerimeter() { 1 usage
23        return 2 * (width + height);
24    }
25
26    public static void main(String[] args) {
27
28        // TODO 5: Create a Rectangle object with sample width and height values.
29        Rectangle rectangle = new Rectangle( width: 8.0, height: 5.0);
30
31        // TODO 6: Print the rectangle's width, height, area, and perimeter.
32        System.out.println("Rectangle Width: " + rectangle.width);
33        System.out.println("Rectangle Height: " + rectangle.height);
34        System.out.println("Area: " + rectangle.calculateArea());
35        System.out.println("Perimeter: " + rectangle.calculatePerimeter());
36    }
37 }
```

Program Rectangle calculates the area and perimeter of a rectangle and outputs the result in the screen. Private double variables are declared at the start of the public class to store the values of the rectangle. A construct in Java is similar to a struct in C++, it will create an object with one or more variables that may be used multiple times with different references using the parameter variables double width, and double height. In the constructor named “Rectangle” the object variables are called attributes and are defined as the global variables width and height.

Public double calculateArea() is a method which calculates the area of the given rectangle and returns a double value. It works similar to a function where it can be defined in the program but the code inside the braces will not evaluate unless the method is called inside the main method. Similarly, *public double calculatePerimeter()* evaluates the perimeter and returns a double value.

For the main method, an object named rectangle of type Rectangle is assigned with width and height as seen in line 29 of the image. Now the variable rectangle with all lowercase has its values and the methods can then be called to calculate area and perimeter. *System.out.println* will output the rectangles width and height using the object’s variables as seen in lines 32 and 33. For area and perimeter the methods are executed with rectangle’s values.

Conclusion

This module introduces basic concepts in Java and the anatomy of the codes structure. A Java program will start with a public class and contains the main method where execution starts. The programs shown before show the use of these basic concepts and offers a brief introduction of Java.

A Java program is defined by a public class with the same name of the program, which will contain the main method that will be the main body of the code. Inside this public class we can define constructors and methods to be used inside the main class.